

Shreyansh Padhy

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SUMMARY

Second-year Electrical and Computer Engineering student at the University of Washington interested in machine learning, hardware engineering and biomedical and hardware research. Experienced in building ML pipelines and sequence modeling for genomics and systems biology, lab quality research and paper writing, and micro-controller engineering.

EDUCATION

University of Washington

Bachelor of Science, Electrical and Computer Engineering

GPA: 3.6

Seattle, WA

2025 – 2029

RELEVANT COURSEWORK

Mathematics: Differential Equations, Linear Algebra, Advanced Multivariable Calculus

Physics & Science: Physics Series (Mechanics, E&M), Statics, Circuit Analysis

EXPERIENCE

IT Support Assistant

2026 – Present

GIX, University of Washington

Seattle, WA

- Designed and built a calendar scheduling system integrated with the Outlook API, automating availability checks and appointment booking for a core team of 15, extending to 70+ additional students.
- Implemented headless multi gpu node cluster for remote processing of tasks for robotics classes using Slurm.
- Built and maintained the team website using Markdown-based content workflows, improving update speed and consistency.
- Managed and delivered presentations for team stakeholders and collaborated cross-functionally to troubleshoot and resolve user-reported technical issues.

Bioinformatics Intern

2025 – Present

Beliveau Lab, University of Washington

Seattle, WA

- Developed deep learning models to predict oligonucleotide hybridization probability (p_{dup}).
- Built a LOCO cross-validation pipeline to evaluate model generalization across the human genome.
- Achieved $R^2 = 0.808$ using a Siamese BiLSTM architecture, outperforming XGBoost by 19%.
- Optimized training for CUDA/MPS and integrated NUPACK thermodynamics as features.

AI Software Developer

2024 – 2025

Brain Capital AI

Remote

- Built backend infrastructure and data pipelines for an AI-powered college advising platform.
- Designed frontend prototypes using Figma to improve user interface usability.

TECHNICAL SKILLS

Programming: Python, Java, C++, SQL, Linux/Bash

ML / Data: PyTorch, scikit-learn, XGBoost, Pandas, NumPy, Sequence Modeling

Tools & Hardware: Git, AutoCAD, Figma, NUPACK, Arduino, GPU Optimization (CUDA, MPS), IoT